

Data Interpretation Exercise

Data Interpretation Assessment (R Programming)

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Topics:

- Mapping Data onto Aesthetics
- Coordinate Systems and Axes
- Color Scales

Instructions: Use the given dataset to write an appropriate R program, generate the required visualization, and answer the questions.

Question 1: Mapping Data onto Aesthetics

Dataset			
Product	Sales (₹ Lakhs)	Profit (%)	Category
P1	120	18	Electronics
P2	90	12	Clothing
P3	150	25	Electronics
P4	80	8	Furniture
P5	135	22	Furniture
P6	100	16	Clothing

1. Write an R program to create a scatter plot with Sales on the x-axis and Profit on the y-axis.

Answer:

```
library(ggplot2)
```

```
sales_data <-
```

```
  data.frame( Product=c("P1","P2","P3","P4","P5","P6"),  
    Sales=c(120,90,150,80,135,100),  
    Profit=c(18,12,25,8,22,16),  
    Category=c("Electronics","Clothing","Electronics","Furniture","Furniture","Clothing")  
  )
```

```
ggplot(sales_data,aes(x=Sales,y=Profit))+  
  geom_point(size=4,color="blue")+  
  labs(title="Sales vs Profit",x="Sales (₹ Lakhs)",y="Profit (%)")+ theme_minimal()
```

2. Map Category using an appropriate aesthetic and Sales using another suitable aesthetic.

Answer:

Answer:

Use Category as color and Sales as size.

```
ggplot(sales_data,aes(x=Sales,y=Profit,color=Category,size=Sales))+ geom_point(alpha=0.8)+  
  
theme_minimal()
```

3. Interpret the relationship between Sales and Profit.

Answer:

Answer:

The relationship is positive. As Sales increase, Profit generally increases. P3 has the highest sales and profit, while P4 has the lowest.

4. Identify the type of scale used for each variable.

Answer:

Answer:

Sales – Continuous

Profit – Continuous

Category – Categorical

5. Suggest one alternative visualization that better represents the same data.

Answer:

Answer:

A Bubble Chart is a suitable alternative because it can display Sales, Profit, Category, and point size together.

Question 2: Coordinate Systems (Cartesian and Polar)

Dataset

Direction	Frequency
N	18
NE	12
E	20
SE	15
S	22
SW	11

W	16
NW	14

1. Write an R program to visualize the dataset using a Cartesian coordinate system.

Answer:

```
library(ggplot2)
df<-data.frame(Direction=c("N","NE","E","SE","S","SW","W","NW"),
Frequency=c(18,12,20,15,22,11,16,14))
ggplot(df,aes(Direction,Frequency))+geom_col(fill="steelblue")+labs(title="Directional
Frequency")+theme_minimal()
```

2. Write another R program to visualize the same data using a Polar coordinate system.

Answer:

```
library(ggplot2)
ggplot(df,aes(Direction,Frequency,fill=Direction))+geom_col()+coord_polar()+theme_minimal()
```

3. Compare the effectiveness of both visualizations.

Answer:

Answer:

The Cartesian chart is easier for comparing frequencies accurately. The Polar chart better represents the circular nature of directional data but makes precise comparisons slightly harder.

4. Explain which coordinate system is more appropriate for directional data.

Answer:

Answer:

The Polar coordinate system is more appropriate because directions are naturally arranged around a circle (0°–360°).

5. Justify your answer based on data interpretation.

Answer:

Answer:

A Polar chart reflects compass directions in their natural positions, making directional patterns easier to interpret, while a Cartesian chart is better for exact value comparisons.

Question 3: Color Scales for Data Representation

Dataset

Student	Marks
A	45
B	58
C	72
D	81
E	67
F	90

1. Write an R program to create a bar chart using a sequential color scale based on Marks.

Answer:

```
library(ggplot2)
df<-data.frame(Student=c("A","B","C","D","E","F"),Marks=c(45,58,72,81,67,90))
ggplot(df,aes(Student,Marks,fill=Marks))+geom_col()+
scale_fill_gradient(low="lightblue",high="darkblue")+
labs(title="Student Marks")+theme_minimal()
```

2. Identify the students with the highest and lowest marks from the visualization.

Answer:

Answer:

Highest: Student F (90)

Lowest: Student A (45)

3. Explain why a sequential color scale is appropriate.

Answer:

Answer:

Marks are continuous numerical values. A sequential color scale shows low-to-high values clearly using gradual color intensity.

4. Suggest a color palette suitable for color-blind users.

Answer:

Answer:

The Viridis color palette is recommended because it is color-blind friendly and easy to

interpret.

R code:

```
+ scale_fill_viridis_c()
```

5. Explain the difference between sequential and categorical color scales.

Answer:

Answer:

Sequential color scales represent ordered numerical data with gradual color changes, while categorical color scales assign distinct colors to different categories without implying any order.

Question 4: Highlighting with Color

Dataset	
Region	Revenue (₹ Crores)
North	820
South	760
East	790
West	1010
Central	770

1. Write an R program to create a bar chart highlighting the region with the highest revenue.

Answer:

```
library(ggplot2)
```

```
revenue_data <-
```

```
  data.frame(Region=c("North","South","East","West","Central"),  
             Revenue=c(820,760,790,1010,770))
```

```
revenue_data$Highlight <- ifelse(revenue_data$Region=="West","Highest","Other")
```

```
ggplot(revenue_data,aes(x=Region,y=Revenue,fill=Highlight))+  
  geom_col()+  
  scale_fill_manual(values=c("Highest"="red","Other"="gray"))+  
  labs(title="Revenue by Region",  
       x="Region",  
       y="Revenue (₹ Crores)")+  
  theme_minimal()
```

2. Explain how highlighting improves data interpretation.

Answer:

Highlighting draws the viewer's attention to the most important data point, making it easier to identify trends, compare values, and quickly recognize the highest-performing region.

3. Identify the highest-performing region.

Answer:

Answer:

The West region has the highest revenue with ₹1010 Crores.

4. Discuss how excessive use of colors can affect interpretation.

Answer:

Answer:

Using too many colors can confuse the viewer, reduce visual clarity, and make it difficult to identify the key information. Important highlights may lose their impact.

5. Suggest one alternative highlighting technique other than color.

Answer:

Answer:

Instead of using color, labels or annotations can be added to highlight the highest value. Increasing the bar size or using a different border style is also an effective technique.

Question 5: Integrated Visualization

Dataset

City	Population (Million)	AQI	Rainfall (mm)	Region
A	8.2	110	850	North
B	5.6	180	430	South
C	9.4	90	1200	North
D	6.8	240	520	East
E	4.9	150	680	West

1. Write an R program to create a bubble chart using appropriate aesthetic mappings.

Answer:

```
library(ggplot2)
```

```
city_data <-
```

```
data.frame( City=c("A","B","  
C","D","E"),
```

```
Population=c(8.2,5.6,9.4,6.8,4.9),
```

```
AQI=c(110,180,90,240,150),
```

```
Rainfall=c(850,430,1200,520,680),
```

```
Region=c("North","South","North","East","West")
)
```

```
ggplot(city_data,
  aes(x=Population,
      y=AQI,
      size=Rainfall,
      color=Region))+
geom_point(alpha=0.8)+
labs(title="Population vs AQI",
x="Population (Million)",
y="AQI")+
theme_minimal()
```

2. Select suitable aesthetics for Population, AQI, Rainfall, and Region.

Answer:

Answer:

Population → X-axis

AQI → Y-axis

Rainfall → Bubble Size

Region → Color

3. Interpret the relationship between Population and AQI.

Answer:

Answer:

The dataset does not show a strong positive relationship. Some cities with higher populations have lower AQI, while others with moderate populations have higher AQI. Therefore, no clear linear relationship is observed.

4. Identify the continuous and categorical variables.

Answer:

Answer:

Continuous Variables:

- Population
- AQI
- Rainfall

Categorical Variables:

- City
- Region

5. Recommend one improvement to enhance the visualization's readability.

Answer:

Answer:

Add labels for each city using `geom_text()` or `geom_label()`, use a color-blind-friendly palette, and include a clear legend to improve readability.